

AI Project Manager

Reads every project card on a Trello board, decides health with deterministic rules, and sends the founder a short report on Telegram.

• OVERDUE • BLOCKED • AT RISK • ON TRACK

12

Trello fields read per card

9

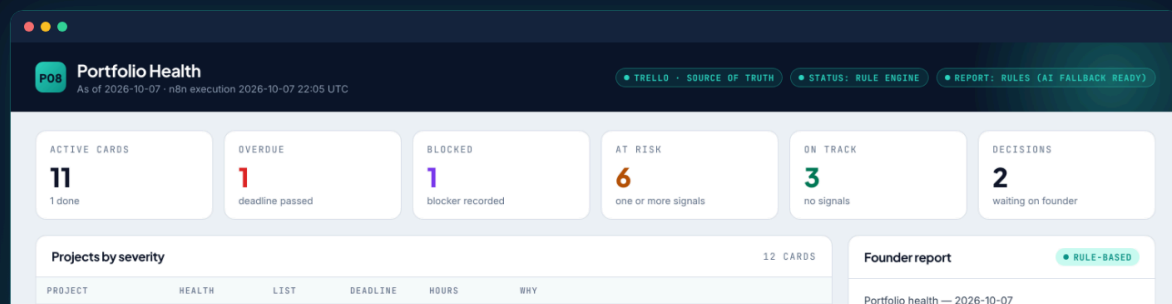
deterministic risk signals

22/22

automated tests passing

1.9 s

full n8n run, Trello → Telegram



P08 · AI Project Manager · Trello + n8n + Gemini + Telegram · Muhammad Rehan Ansar

2026

CASE STUDY

Automated project health reporting for a founder, built on Trello and n8n

A Trello board is the source of truth. An n8n workflow reads it every weekday morning, a deterministic rule engine decides the health of each project, Gemini turns the facts into a short founder note, and Telegram delivers it. If the AI fails or contradicts the rules, the rule-based report is sent instead.

Status lives in Trello, but nobody reads every card

Teams track delivery in Trello. The founder needs to know which projects are late, blocked or drifting, but the signals are spread across due dates, custom fields, card activity and links between cards. Checking manually takes time and is easy to skip, so problems surface late.

Asking an AI model "which projects are at risk?" is not a fix: the answer can change between runs and cannot be audited.

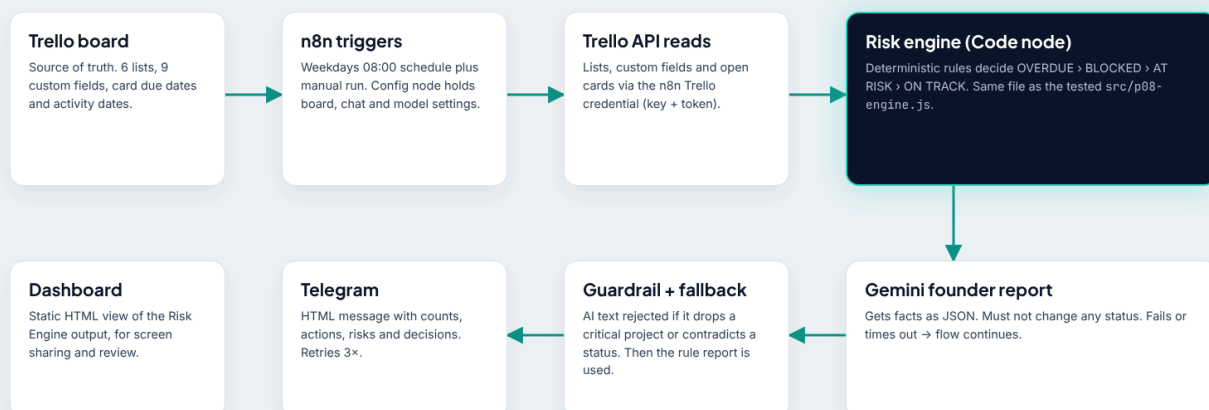
Rules decide. AI explains. n8n delivers.

- **Trello** — 6 lists (Backlog, To Do, In Progress, Blocked, Review, Done) and 9 custom fields: Owner, Priority, Estimated Hours, Actual Hours, Blocker, Dependency, Risk Level, Next Milestone, Decision Required. Project = card name, Status = list, Deadline = due date.
- **n8n** — schedule (weekdays 08:00) and manual triggers; three Trello API reads using the n8n Trello credential; Code nodes; Gemini HTTP call; Telegram node.
- **Risk engine** — one dependency-free JavaScript file, unit tested, embedded unchanged into the n8n Code node by a build script.
- **Gemini** — receives only computed facts as JSON with instructions not to change any status. Output is checked by a guardrail before use.
- **Telegram** — HTML message with counts, actions, risks and founder decisions.

ARCHITECTURE

Trello → n8n → rules → Gemini → Telegram

Each layer does one job. Trello holds the data, n8n runs the flow, the rule engine decides health, Gemini only writes the summary, and Telegram delivers it.



• Credentials live only in n8n • No secrets in git • AI never sets status • Report always delivered

10 nodes, executed end to end

EXECUTION

Successful run in 1.9 seconds

Real n8n execution: 6 lists → 9 custom fields → 12 cards → 1 report → Telegram. Earlier failed runs stay visible in the history.

• SUCCEEDED • EXECUTION #3

The AI can only rephrase

The Select Report node accepts Gemini's text only if it is non-empty, under the length limit, names every OVERDUE and BLOCKED project, and never calls a non-healthy project "on track". Otherwise the rule-based report is sent. All three paths were executed in n8n:

GEMINI RESPONSE	REPORT SENT	RESULT
Request failed (HTTP 400)	Rule-based	Pass — delivered to Telegram
Faithful summary	Gemini	Pass
Calls an AT RISK project "on track"	Rule-based	Pass — contradiction rejected

What the founder receives

TELEGRAM

The founder gets one clear message

Exact text sent by the n8n Telegram node (message accepted by the Telegram Bot API, ok = true). Gemini was unavailable in this run, so the guardrail used the rule-based report.

• DELIVERED • AI FALLBACK PATH

P08 Project Health — 2026-10-07

● Overdue 1 ● Blocked 1 ● At risk 6 ● On track 3

Status decided by rules · report by rule engine (AI fallback)

Portfolio health — 2026-10-07

11 active: 1 overdue, 1 blocked, 6 at risk, 3 on track. 1 done.

Needs action now:

- Client Portal v2 [OVERDUE] (Sara Malik): Deadline 2026-10-02 passed 5 day(s) ago
- Payment Gateway Migration [BLOCKED] (Omar Haddad): Blocker: Waiting for bank sandbox credentials

At risk:

- Security Audit Remediation (Khalid Rahman): Due in 5 day(s) but still in "To Do"
- Mobile App Release 3.4 (Lina Farouk): Depends on "Payment Gateway Migration", which is BLOCKED
- Support Chatbot Rollout (Yusuf Amin): Depends on "Knowledge Base Cleanup" (due 2026-11-10), which finishes after this deadline (2026-10-30)
- Data Warehouse Sync (Bilal Qureshi): 55h logged vs 40h estimated (+38%)
- Customer Onboarding Emails (Hana Yusuf): No update for 15 days
- Internal Analytics Dashboard (no owner): Missing: Owner

Decisions required from you:

- Client Portal v2: Approve a 1-week scope cut or one extra developer
- Security Audit Remediation: Approve the external pen-test vendor

On track: CRM Data Import, Knowledge Base Cleanup, Marketing Website Refresh

No credentials in code

- Trello key + token, Gemini key and Telegram bot token are stored as n8n credentials. The workflow JSON references them by name only.
- 22/22 automated tests pass (rules, edge cases, invalid input, Trello parsing, guardrail, Telegram escaping and length).
- Verified: full n8n run with Trello API responses served from a local test server, real Telegram delivery, all three report paths.
- Not verified in this build environment: a live Trello board and a live Gemini response. Both use the same nodes and only need credentials.